

Exploratory Data Analysis (EDA) for EuroSAT Dataset

1. Introduction

The EuroSAT dataset is a satellite image dataset used for land-use and land-cover classification tasks. It contains RGB satellite images categorized into 10 different classes representing various types of land cover. The purpose of this EDA is to understand the dataset structure, inspect image characteristics, analyze class distribution, and verify data quality before model training.

2. Dataset Overview

The dataset contains 27,000 images distributed across 10 classes:

- AnnualCrop
- Forest
- HerbaceousVegetation
- Highway
- Industrial
- Pasture
- PermanentCrop
- Residential
- River
- SeaLake

3. Class Distribution Analysis

The number of images in each class was analyzed to determine whether the dataset suffers from class imbalance.

Class	Number of Images
AnnualCrop	3000
Forest	3000
HerbaceousVegetation	3000
Highway	2500
Industrial	2500
Pasture	2000
PermanentCrop	2500
Residential	3000
River	2500

Class	Number of Images
SeaLake	3000

The dataset is relatively balanced, with image counts ranging from 2,000 to 3,000 images per class.

4. Visual Inspection

Sample images from each class were displayed and inspected. The classes show noticeable differences in texture, color, and spatial patterns. For example, forests contain dense vegetation, residential areas contain clusters of buildings, and rivers contain elongated water bodies.

5. Image Properties

The analysis showed that:

- All images have a fixed resolution of 64×64 pixels.
- All images are stored in RGB format.
- No grayscale images were detected.

This consistency simplifies preprocessing and model training.

6. Data Quality Assessment

A corrupted image check was performed on the entire dataset.

Result:

- Corrupted Images: 0

This indicates that the dataset is clean and suitable for deep learning applications.

7. Conclusion

The EuroSAT dataset contains 27,000 RGB satellite images distributed across 10 classes. The dataset is relatively balanced, all images have uniform dimensions (64×64), and no corrupted images were found. Based on the exploratory analysis, the dataset is well-structured and suitable for image classification tasks using deep learning models.